

Recursive Macroeconomic Theory: Chapters 4–6

Notes with Derivations

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1 Unifying intuition across Chapters 4–6

The three chapters all use the same recursive logic:

$$V(\text{state today}) = \max_{\text{choice today}} \{ \text{current payoff} + \beta \mathbb{E}[V(\text{state tomorrow})] \}.$$

The chapters differ mainly in what the choice is.

Chapter	Central choice	Main mathematical object
Chapter 4	How much to adjust a dynamic system	Linear feedback rule $u = -Fx$
Chapter 5	Accept now or keep searching	Reservation wage $\bar{w}$
Chapter 6	Do individual rules aggregate consistently?	Fixed point / equilibrium law

Chapter 4 is a theory of *control*. The state summarizes everything relevant from the past; the control moves the state; the value of changing the state is summarized by a quadratic value function. Chapter 5 is a theory of *option value*. Rejecting an offer preserves the possibility of a better future draw, so optimal behavior is usually a cutoff rule. Chapter 6 adds *equilibrium consistency*. Agents optimize while taking prices or aggregate laws as given, but in equilibrium the laws generated by their policies must coincide with the laws they believed.

2 Chapter 4: Linear quadratic dynamic programming

Exercise 4.1: Modified linear regulator with a cross term

Consider

$$\max_{\{u_t\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t (x_t' R x_t + u_t' Q u_t + 2u_t' H x_t), \quad x_{t+1} = A x_t + B u_t.$$

Assume the matrices are conformable and that the quadratic maximization is concave in u .

Guess a quadratic value function

$$V(x) = x' P x.$$

The Bellman equation is

$$\begin{aligned} V(x) &= \max_u \{ x' R x + u' Q u + 2u' H x + \beta (A x + B u)' P (A x + B u) \} \\ &= \max_u \{ x' R x + u' Q u + 2u' H x + \beta x' A' P A x + 2\beta u' B' P A x + \beta u' B' P B u \}. \end{aligned}$$

Collect terms involving u :

$$V(x) = \max_u \{ x' (R + \beta A' P A) x + u' (Q + \beta B' P B) u + 2u' (H + \beta B' P A) x \}.$$

Define

$$M(P) = Q + \beta B' P B, \quad N(P) = H + \beta B' P A.$$

Then

$$V(x) = \max_u \{ x' (R + \beta A' P A) x + u' M u + 2u' N x \}.$$

The first-order condition is

$$2M u + 2N x = 0,$$

so the optimal control is

$$u^*(x) = -M(P)^{-1}N(P)x$$

or, equivalently,

$$u_t = -(Q + \beta B'PB)^{-1}(\beta B'PA + H)x_t.$$

To obtain the Riccati equation, complete the square:

$$u'Mu + 2u'Nx = (u + M^{-1}Nx)'M(u + M^{-1}Nx) - x'N'M^{-1}Nx.$$

At the optimum, the square term is zero. Therefore,

$$V(x) = x' [R + \beta A'PA - N'M^{-1}N] x.$$

Because $V(x) = x'Px$ for all x , the matrix P must satisfy

$$P = R + \beta A'PA - (\beta A'PB + H')(Q + \beta B'PB)^{-1}(\beta B'PA + H).$$

A simple value-iteration algorithm for P is

$$P_{j+1} = \mathcal{T}(P_j),$$

where

$$\mathcal{T}(P) = R + \beta A'PA - (\beta A'PB + H')(Q + \beta B'PB)^{-1}(\beta B'PA + H).$$

Starting from $P_0 = 0$, iterate until

$$\|P_{j+1} - P_j\| < \varepsilon.$$

Then compute the feedback matrix

$$F = (Q + \beta B'PB)^{-1}(\beta B'PA + H), \quad u_t = -Fx_t.$$

Exercise 4.2: Policy improvement for the discounted regulator

The idea of policy improvement is to alternate between two steps: evaluate a fixed policy, then improve it.

Suppose the current policy is

$$u_t = -F_j x_t.$$

The closed-loop law of motion is

$$x_{t+1} = (A - BF_j)x_t.$$

For the standard LQ payoff $x'Rx + u'Qu$, the one-period return under this fixed policy is

$$x'_t R x_t + (-F_j x_t)' Q (-F_j x_t) = x'_t (R + F'_j Q F_j) x_t.$$

If the value of using F_j forever is

$$V_j(x) = x' P_j x,$$

then P_j solves the Lyapunov equation

$$P_j = R + F'_j Q F_j + \beta (A - BF_j)' P_j (A - BF_j).$$

This is the *policy-evaluation* step.

For policy improvement, take P_j as given and solve the one-step problem

$$\max_u \{x'Rx + u'Qu + \beta(Ax + Bu)'P_j(Ax + Bu)\}.$$

The first-order condition is

$$(Q + \beta B'P_jB)u + \beta B'P_jAx = 0.$$

Thus the improved policy is

$$u = -F_{j+1}x,$$

where

$$F_{j+1} = (Q + \beta B'P_jB)^{-1}\beta B'P_jA.$$

The algorithm is:

$$F_j \longrightarrow P_j \longrightarrow F_{j+1}.$$

At convergence, $F_{j+1} = F_j = F$, and the pair (P, F) solves the usual Riccati system.

Exercise 4.3: Household saving and investment as an LQ problem

The household maximizes

$$-\sum_{t=1}^{\infty} \beta^t [(c_t - b)^2 + \gamma i_t^2]$$

subject to

$$c_t + i_t = ra_t + y_t, \quad a_{t+1} = a_t + i_t, \quad y_{t+1} = \rho_1 y_t + \rho_2 y_{t-1}.$$

Choose the control

$$u_t = i_t$$

and the state

$$x_t = \begin{bmatrix} a_t \\ y_t \\ y_{t-1} \\ 1 \end{bmatrix}.$$

The constant 1 is included because the target b creates linear and constant terms in the quadratic payoff.

The state evolves as

$$x_{t+1} = Ax_t + Bu_t,$$

where

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \rho_1 & \rho_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Consumption is

$$c_t = ra_t + y_t - i_t.$$

Define

$$d = \begin{bmatrix} r \\ 1 \\ 0 \\ -b \end{bmatrix}, \quad d'x_t = ra_t + y_t - b.$$

Then

$$c_t - b = d'x_t - u_t.$$

The one-period return is

$$\begin{aligned} -[(c_t - b)^2 + \gamma u_t^2] &= -[(d'x_t - u_t)^2 + \gamma u_t^2] \\ &= -x_t' d d' x_t + 2u_t d' x_t - (1 + \gamma)u_t^2. \end{aligned}$$

Thus, in LQ form,

$$x_t' R x_t + u_t' Q u_t + 2u_t' H x_t,$$

we can use

$$\boxed{R = -dd', \quad Q = -(1 + \gamma), \quad H = d'.$$

The optimal policy is

$$u_t = -F x_t.$$

For the parameter values

$$(\beta, 1 + r, b, \gamma, \rho_1, \rho_2) = (.95, .95^{-1}, 30, 1, 1.2, -.3),$$

the reported feedback rule is approximately

$$\boxed{F = [-0.0095, -0.8127, -0.0488, 5.4423].}$$

Therefore

$$i_t = -F \begin{bmatrix} a_t \\ y_t \\ y_{t-1} \\ 1 \end{bmatrix}.$$

The rule says that investment responds to assets, current labor income, lagged labor income, and the target consumption level.

Exercise 4.4: Durables and habits

Now utility depends on consumption services

$$s_t = \lambda h_t + \pi c_t,$$

and the stock h_t evolves according to

$$h_{t+1} = \delta h_t + \theta c_t.$$

The objective is

$$-\sum_{t=1}^{\infty} \beta^t [(s_t - b)^2 + \gamma i_t^2].$$

Again set $u_t = i_t$ and use

$$c_t = r a_t + y_t - u_t.$$

The state must now include h_t :

$$x_t = \begin{bmatrix} a_t \\ y_t \\ y_{t-1} \\ h_t \\ 1 \end{bmatrix}.$$

The laws of motion are

$$\begin{aligned} a_{t+1} &= a_t + u_t, \\ y_{t+1} &= \rho_1 y_t + \rho_2 y_{t-1}, \\ h_{t+1} &= \delta h_t + \theta(ra_t + y_t - u_t). \end{aligned}$$

Therefore

$$x_{t+1} = Ax_t + Bu_t,$$

with

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \rho_1 & \rho_2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \theta r & \theta & 0 & \delta & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -\theta \\ 0 \end{bmatrix}.$$

Services satisfy

$$\begin{aligned} s_t - b &= \lambda h_t + \pi(ra_t + y_t - u_t) - b \\ &= Dx_t - \pi u_t, \end{aligned}$$

where

$$D = \begin{bmatrix} \pi r & \pi & 0 & \lambda & -b \end{bmatrix}.$$

Thus

$$\begin{aligned} -[(s_t - b)^2 + \gamma u_t^2] &= -[(Dx_t - \pi u_t)^2 + \gamma u_t^2] \\ &= -x_t' D' D x_t + 2u_t(\pi D)x_t - (\pi^2 + \gamma)u_t^2. \end{aligned}$$

So one valid LQ representation is

$$\boxed{R = -D' D, \quad Q = -(\pi^2 + \gamma), \quad H = \pi D.}$$

For

$$(\lambda, \pi, \delta, \theta) = (1, .05, .95, 1),$$

we have

$$h_{t+1} = .95h_t + c_t, \quad s_t = h_t + .05c_t.$$

This is naturally interpreted as durable consumption: the stock produces most services, while current purchases provide only a small immediate service flow.

For

$$(\lambda, \pi, \delta, \theta) = (-1, 1, .95, 1),$$

we have

$$h_{t+1} = .95h_t + c_t, \quad s_t = c_t - h_t.$$

This is naturally interpreted as habit persistence: utility depends on current consumption relative to the habit stock.

Exercise 4.5: Forecasting a discounted sum of future labor income

Labor income follows

$$y_{t+1} = \rho_1 y_t + \rho_2 y_{t-1} + w_{t+1} + \gamma w_t.$$

Equivalently,

$$y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + w_t + \gamma w_{t-1}.$$

Using the lag operator L ,

$$(1 - \rho_1 L - \rho_2 L^2) y_t = (1 + \gamma L) w_t.$$

Hence

$$y_t = c(L) w_t, \quad c(L) = \frac{1 + \gamma L}{1 - \rho_1 L - \rho_2 L^2}.$$

We want

$$S_t = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j y_{t+j}.$$

Let

$$c(L) = \sum_{k=0}^{\infty} c_k L^k.$$

Then

$$y_{t+j} = \sum_{k=0}^{\infty} c_k w_{t+j-k}.$$

Conditional on information at t , only shocks dated t or earlier are known. Therefore

$$\mathbb{E}_t y_{t+j} = \sum_{k=j}^{\infty} c_k w_{t+j-k}.$$

So

$$S_t = \sum_{j=0}^{\infty} \beta^j \sum_{k=j}^{\infty} c_k w_{t+j-k}.$$

Let $m = k - j$, so $k = m + j$. Then

$$S_t = \sum_{m=0}^{\infty} \left(\sum_{j=0}^{\infty} \beta^j c_{m+j} \right) w_{t-m}.$$

Thus the discounted forecast is another one-sided linear filter in current and past shocks. A compact expression for that filter is

$$S_t = \left[\frac{c(L) - \beta c(\beta) L^{-1}}{1 - \beta L^{-1}} \right] w_t.$$

The identity follows because if $d(L) = \sum_{m \geq 0} d_m L^m$ with

$$d_m = \sum_{j=0}^{\infty} \beta^j c_{m+j},$$

then

$$d(L)(1 - \beta L^{-1}) = c(L) - \beta d_0 L^{-1} = c(L) - \beta c(\beta) L^{-1}.$$

Substituting the ARMA transfer function gives

$$S_t = \frac{L(1 + \gamma L) - \beta c(\beta)(1 - \rho_1 L - \rho_2 L^2)}{(1 - \rho_1 L - \rho_2 L^2)(L - \beta)} w_t.$$

This formula is the object that a program computes when it plots impulse responses, spectra, autocovariances, and simulations for

$$(\beta, \rho_1, \rho_2, \gamma) = (.95, 1.2, -.4, .5).$$

Exercise 4.6: Dynamic Laffer curves

Money demand is

$$\frac{M_t}{p_t} = \gamma_1 - \gamma_2 \frac{p_{t+1}}{p_t},$$

so

$$M_t = \gamma_1 p_t - \gamma_2 p_{t+1}.$$

The government budget constraint is

$$\frac{M_t - M_{t-1}}{p_t} = g,$$

so

$$M_t = M_{t-1} + g p_t.$$

Substitute this into money demand:

$$M_{t-1} + g p_t = \gamma_1 p_t - \gamma_2 p_{t+1}.$$

Hence

$$p_{t+1} = -\frac{1}{\gamma_2} M_{t-1} + \frac{\gamma_1 - g}{\gamma_2} p_t.$$

Together with $M_t = M_{t-1} + g p_t$,

$$\begin{bmatrix} M_t \\ p_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & g \\ -1/\gamma_2 & (\gamma_1 - g)/\gamma_2 \end{bmatrix} \begin{bmatrix} M_{t-1} \\ p_t \end{bmatrix}.$$

For

$$\gamma_1 = 100, \quad \gamma_2 = 50, \quad g = .05,$$

the transition matrix is

$$A = \begin{bmatrix} 1 & .05 \\ -.02 & 1.999 \end{bmatrix}.$$

Its two eigenvalues are approximately

$$\lambda_1 \approx 1.001, \quad \lambda_2 \approx 1.998.$$

A general path can be written as

$$\begin{bmatrix} M_{t-1} \\ p_t \end{bmatrix} = \alpha \lambda_1^t v_1 + \eta \lambda_2^t v_2.$$

The initial nominal money stock M_{-1} is fixed, but the initial price p_0 is not pinned down by a stable root because both roots exceed one. Hence there is a continuum of perfect-foresight equilibria indexed by p_0 , subject to positivity of money and prices.

The minimum admissible p_0 selects the low-inflation path associated with λ_1 . Other admissible paths eventually become dominated by the larger root λ_2 .

The Laffer-curve logic is clearest in a stationary gross-inflation path with

$$\frac{p_{t+1}}{p_t} = \frac{M_t}{M_{t-1}} = \lambda.$$

Real balances are

$$\frac{M_t}{p_t} = \gamma_1 - \text{gamma}_2 \lambda.$$

Seigniorage revenue is

$$\begin{aligned} g &= \frac{M_t - M_{t-1}}{p_t} \\ &= \left(1 - \frac{1}{\lambda}\right) \frac{M_t}{p_t} \\ &= \left(1 - \frac{1}{\lambda}\right) (\gamma_1 - \text{gamma}_2 \lambda). \end{aligned}$$

Thus the same g can correspond to two inflation rates: a low-inflation equilibrium with a large real money base, and a high-inflation equilibrium with a small real money base.

3 Chapter 5: Search, matching, and unemployment

A useful reservation-wage lemma

Many exercises in Chapter 5 have the form

$$V(w) = \max\{A(w), C\},$$

where $A(w)$ is increasing in w and C does not depend on w . Then the optimal policy is a cutoff rule:

$$\text{accept if } w \geq \bar{w}, \quad \text{reject if } w < \bar{w},$$

where

$$A(\bar{w}) = C.$$

If accepting a permanent wage w gives $w/(1 - \beta)$, then

$$V(w) = \frac{1}{1 - \beta} \max\{w, \bar{w}\}.$$

For a wage distribution G with upper bound B ,

$$\mathbb{E}_G \max\{W, \bar{w}\} = \bar{w} + \int_{\bar{w}}^B [1 - G(s)] ds.$$

This identity follows from integration by parts:

$$\int_{\bar{w}}^B w dG(w) = B - \bar{w}G(\bar{w}) - \int_{\bar{w}}^B G(s) ds,$$

so

$$\bar{w}G(\bar{w}) + \int_{\bar{w}}^B w dG(w) = \bar{w} + \int_{\bar{w}}^B [1 - G(s)] ds.$$

Exercise 5.1: Chance of no offer

With probability ϕ the worker receives no offer; with probability $1 - \phi$ she receives a wage draw from F . The Bellman equation is

$$V(w) = \max \left\{ \frac{w}{1 - \beta}, c + \phi\beta V(0) + (1 - \phi)\beta \int V(w') dF(w') \right\}.$$

The acceptance value $w/(1 - \beta)$ is increasing in w , while the rejection value is independent of w . Therefore there is a reservation wage $\bar{w}$.

At the cutoff,

$$\frac{\bar{w}}{1 - \beta} = c + \phi\beta V(0) + (1 - \phi)\beta \mathbb{E}V(W').$$

If $\bar{w} > 0$, then a zero offer is rejected and

$$V(0) = \frac{\bar{w}}{1 - \beta}.$$

Also,

$$\mathbb{E}V(W') = \frac{1}{1 - \beta} \mathbb{E} \max\{W', \bar{w}\}.$$

Multiplying the indifference condition by $1 - \beta$ gives

$$\bar{w} = (1 - \beta)c + \beta\phi\bar{w} + \beta(1 - \phi)\mathbb{E} \max\{W', \bar{w}\}.$$

Using

$$\mathbb{E} \max\{W', \bar{w}\} = \bar{w} + \int_{\bar{w}}^B [1 - F(s)] ds,$$

we obtain

$$(1 - \beta)\bar{w} = (1 - \beta)c + \beta(1 - \phi) \int_{\bar{w}}^B [1 - F(s)] ds.$$

No-offer risk lowers the option value of waiting because the worker receives fewer future opportunities.

Exercise 5.2: Two offers per period

Let W_1, W_2 be independent draws from F . The worker only cares about

$$W^{(2)} = \max\{W_1, W_2\}.$$

The distribution of the maximum is

$$G_2(w) = \Pr(W^{(2)} \leq w) = \Pr(W_1 \leq w, W_2 \leq w) = F(w)^2.$$

Therefore

$$V(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta \int V(w') dF(w')^2 \right\}.$$

The reservation wage $\bar{w}_2$ satisfies

$$\bar{w}_2 = (1 - \beta)c + \beta\bar{w}_2 + \beta \int_{\bar{w}_2}^B [1 - F(s)^2] ds.$$

Equivalently,

$$h_2(\bar{w}_2) \equiv (1 - \beta)\bar{w}_2 - \beta \int_{\bar{w}_2}^B [1 - F(s)^2] ds - (1 - \beta)c = 0.$$

For one offer, the cutoff $\bar{w}_1$ satisfies

$$h_1(\bar{w}_1) \equiv (1 - \beta)\bar{w}_1 - \beta \int_{\bar{w}_1}^B [1 - F(s)] ds - (1 - \beta)c = 0.$$

Since

$$F(s)^2 \leq F(s),$$

we have

$$1 - F(s)^2 \geq 1 - F(s).$$

Thus the continuation value of search is larger with two offers. To make the worker indifferent between accepting and rejecting, the current wage must be higher:

$$\bar{w}_2 \geq \bar{w}_1.$$

Exercise 5.3: Random number of offers

Suppose the number of offers n is random, with

$$\Pr(n) = \pi_n, \quad n = 1, \dots, N.$$

Conditional on receiving n offers, the best offer has distribution

$$G_n(w) = F(w)^n.$$

The unconditional distribution of next period's best offer is the mixture

$$G(w) = \sum_{n=1}^N \pi_n F(w)^n.$$

The Bellman equation is

$$V(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta \sum_{n=1}^N \pi_n \int V(w') dF(w')^n \right\}.$$

The reservation wage satisfies

$$(1 - \beta)\bar{w} = (1 - \beta)c + \beta \sum_{n=1}^N \pi_n \int_{\bar{w}}^B [1 - F(s)^n] ds.$$

More probability on larger n raises the option value of waiting and therefore raises the reservation wage.

Exercise 5.4: Markov number of offers

Now the number of offers follows a Markov chain:

$$\Pr(n_{t+1} = m \mid n_t = n) = \pi_{mn}.$$

The state is (w, n) , where w is the best current offer and n is the current number of offers. The Bellman equation is

$$V(w, n) = \max \left\{ \frac{w}{1 - \beta}, c + \beta \sum_{m=1}^N \pi_{mn} \int V(w', m) dF(w')^m \right\}.$$

For each n , the rejection value depends on n but not on w . Therefore the optimal policy is a state-dependent cutoff:

$$\text{accept if } w \geq \bar{w}(n).$$

At the cutoff,

$$\frac{\bar{w}(n)}{1 - \beta} = c + \beta \sum_{m=1}^N \pi_{mn} \int V(w', m) dF(w')^m.$$

Since

$$V(w', m) = \frac{1}{1 - \beta} \max\{w', \bar{w}(m)\},$$

we can write

$$\bar{w}(n) = (1 - \beta)c + \beta \sum_{m=1}^N \pi_{mn} \mathbb{E}_{F^m} \max\{W', \bar{w}(m)\}.$$

Thus n matters because it forecasts the future search environment.

Exercise 5.5: Choosing the number of offers

Let Q be the value before choosing how many offers to solicit. If the worker chooses n , she pays cost $k(n)$ and receives the best of n draws, whose distribution is F^n .

If she accepts the best wage w , current income net of search cost is $w - k(n)$, and from next period onward she receives w . Hence the accept value is

$$w - k(n) + \beta \frac{w}{1 - \beta} = \frac{w}{1 - \beta} - k(n).$$

If she rejects, she receives unemployment compensation net of search cost and returns to the pre-search state next period:

$$c - k(n) + \beta Q.$$

Therefore

$$Q = \max_n \int \max \left\{ \frac{w}{1 - \beta} - k(n), c - k(n) + \beta Q \right\} dF(w)^n.$$

The worker chooses n by equating the marginal value of a better maximum draw with the marginal search cost. Higher n shifts the best-offer distribution upward, but the cost $k(n)$ lowers the payoff in both acceptance and rejection states.

Exercise 5.6: Mortensen externality

There are two sides of a match. Each side chooses search resources. The number of possible matches is

$$n = f(c_1 + c_2).$$

For side i , taking c_j as given, the Bellman equation is

$$Q_i = \max_{c_i} \int \max \left\{ \frac{\theta}{1-\beta} - c_i, -c_i + \beta Q_i \right\} dF(\theta)^{f(c_i+c_j)}.$$

The reservation match quality satisfies

$$\frac{\bar{\theta}_i}{1-\beta} = \beta Q_i.$$

In a Nash equilibrium, each agent internalizes how c_i improves his own chance of drawing a good match, but not how c_i improves the other side's matching opportunities.

A planner with Pareto weight λ solves

$$Q^P(\lambda) = \max_{c_1, c_2} \int \max \left\{ \frac{\theta}{1-\beta} - \lambda c_1 - (1-\lambda)c_2, -\lambda c_1 - (1-\lambda)c_2 + \beta Q^P(\lambda) \right\} dF(\theta)^{f(c_1+c_2)}.$$

The planner internalizes that c_1 and c_2 jointly raise the number of matching opportunities. Thus, when f is increasing, decentralized Nash search effort is generally too low relative to the social optimum.

Exercise 5.7: Variable labor supply

If employed at wage w , the worker chooses leisure $l \in [0, 1]$ and works $1-l$. Income is

$$y = w(1-l).$$

The employed value solves

$$V_E(w) = \max_l \{u(w(1-l), l) + \beta V_E(w)\}.$$

Therefore

$$V_E(w) = \frac{1}{1-\beta} \max_l u(w(1-l), l).$$

Let

$$l^*(w) \in \arg \max_l u(w(1-l), l).$$

Because w is permanent after acceptance, the same $l^*(w)$ is chosen every period.

The unemployed value is

$$V_U(w) = \max \left\{ V_E(w), u(c, 1) + \beta \int V_U(w') dF(w') \right\}.$$

The first term is increasing in w because a worker with a higher wage can always choose the same hours as at a lower wage and earn more income. The second term is independent of w . Hence the optimal policy is a reservation wage rule.

Exercise 5.8: Wage growth and the reservation wage

If the worker accepts initial wage w , wages evolve as

$$w, \phi w, \phi^2 w, \dots$$

Assuming $\beta\phi < 1$, the value of accepting is

$$\sum_{j=0}^{\infty} \beta^j \phi^j w = \frac{w}{1 - \beta\phi}.$$

The Bellman equation is

$$V(w) = \max \left\{ \frac{w}{1 - \beta\phi}, c + \beta \int V(w') dF(w') \right\}.$$

At the reservation wage,

$$\frac{\bar{w}}{1 - \beta\phi} = c + \beta \int V(w') dF(w').$$

Since

$$V(w') = \frac{1}{1 - \beta\phi} \max\{w', \bar{w}\},$$

multiplying by $1 - \beta\phi$ gives

$$\bar{w} = (1 - \beta\phi)c + \beta \mathbb{E} \max\{W', \bar{w}\}.$$

Using the integration identity,

$$\bar{w} = (1 - \beta\phi)c + \beta \bar{w} + \beta \int_{\bar{w}}^B [1 - F(s)] ds.$$

Equivalently,

$$(1 - \beta)\bar{w} - \beta \int_{\bar{w}}^B [1 - F(s)] ds = (1 - \beta\phi)c.$$

The left side is increasing in $\bar{w}$, while the right side decreases as ϕ rises. Therefore a higher wage-growth rate lowers the starting reservation wage:

$$\phi \uparrow \Rightarrow \bar{w} \downarrow.$$

The intuition is that a low starting wage becomes more attractive when it leads to faster future wage growth.

Exercise 5.9: Finite horizon search

The worker lives for two periods. In the last period, accepting yields w and rejecting yields c , so

$$V_2(w) = \max\{w, c\}.$$

The second-period reservation wage is

$$\bar{w}_2 = c.$$

In the first period, accepting a permanent job yields

$$w + \beta w = w(1 + \beta).$$

Rejecting yields

$$c + \beta \int V_2(w') dF(w').$$

Thus

$$V_1(w) = \max \left\{ w(1 + \beta), c + \beta \int V_2(w') dF(w') \right\}.$$

The first-period reservation wage solves

$$\boxed{\bar{w}_1(1 + \beta) = c + \beta \mathbb{E}[V_2(W')].}$$

Since

$$V_2(W') = \max\{W', c\} \geq c,$$

we obtain

$$\bar{w}_1(1 + \beta) \geq c + \beta c = (1 + \beta)c.$$

Therefore

$$\boxed{\bar{w}_1 \geq c = \bar{w}_2.}$$

Reservation wages fall as the terminal date approaches because the option value of waiting declines.

Exercise 5.10: Finite horizon and mean-preserving spreads

Let F_1 be a mean-preserving spread of F_2 . This means both distributions have the same mean, but F_1 is riskier in the convex-order sense:

$$\int \varphi(w) dF_1(w) \geq \int \varphi(w) dF_2(w)$$

for every convex function φ .

Without unemployment compensation, at the final date T ,

$$V_T^i(w) = w,$$

so the worker accepts every wage and

$$\bar{w}_T^1 = \bar{w}_T^2 = 0.$$

At date $T - 1$,

$$V_{T-1}^i(w) = \max \{ w(1 + \beta), \beta \mathbb{E}_i[W'] \}.$$

Because F_1 and F_2 have the same mean,

$$\mathbb{E}_1[W'] = \mathbb{E}_2[W'],$$

so

$$\boxed{\bar{w}_{T-1}^1 = \bar{w}_{T-1}^2.}$$

For earlier dates, the future value function is convex because it is the maximum of an increasing linear acceptance value and a continuation value. A mean-preserving spread raises the expected value of a convex function, so

$$\mathbb{E}_1[V_{t+1}(W')] \geq \mathbb{E}_2[V_{t+1}(W')].$$

This raises the continuation value of search and therefore raises the reservation wage under F_1 for $t \leq T - 2$.

With unemployment compensation c , the terminal value is

$$V_T^i(w) = \max\{w, c\}.$$

This is convex in w . Therefore

$$\mathbb{E}_1[V_T(W')] \geq \mathbb{E}_2[V_T(W')].$$

At $T - 1$ the reservation wage is proportional to this expected continuation value, so the riskier distribution already gives a higher reservation wage at $T - 1$.

Exercise 5.11: Taxes and variable search intensity

Let e be search effort. An offer arrives with probability $\pi(e)$, where

$$\pi'(e) > 0, \quad \pi''(e) < 0.$$

If employed at wage w , the worker earns after-tax income $w - T(w)$ and is fired with probability α .

Let Q be the value of being unemployed before choosing effort. The employed value satisfies

$$E(w) = w - T(w) + \beta(1 - \alpha)E(w) + \beta\alpha Q.$$

Therefore

$$E(w) = \frac{w - T(w)}{1 - \beta(1 - \alpha)} + \frac{\beta\alpha Q}{1 - \beta(1 - \alpha)}.$$

If the worker has searched with effort e and has an offer w , accepting gives

$$A(w, e) = \frac{w - T(w)}{1 - \beta(1 - \alpha)} - e + \frac{\beta\alpha Q}{1 - \beta(1 - \alpha)}.$$

Rejecting gives

$$R(e) = 1 - e + z + \beta Q.$$

The effort cost e appears in both terms and cancels from the accept/reject comparison. Hence the reservation wage is independent of e . The cutoff solves

$$\bar{w} - T(\bar{w}) = [1 - \beta(1 - \alpha)](1 + z + \beta Q) - \beta\alpha Q.$$

Define the after-tax surplus from accepting w rather than the reservation wage:

$$\Delta Y(w) = [w - T(w)] - [\bar{w} - T(\bar{w})].$$

The expected gain from receiving an offer is

$$\frac{1}{1 - \beta(1 - \alpha)} \int_{\bar{w}}^{\infty} \Delta Y(w) dF(w).$$

The effort problem has the form

$$\max_e \left\{ \pi(e) \frac{1}{1 - \beta(1 - \alpha)} \int_{\bar{w}}^{\infty} \Delta Y(w) dF(w) - e \right\}.$$

For an interior optimum,

$$\pi'(e) \frac{1}{1 - \beta(1 - \alpha)} \int_{\bar{w}}^{\infty} \Delta Y(w) dF(w) = 1.$$

For the tax rule

$$T(w) = t(w - a),$$

a larger a raises after-tax income at every wage and is especially favorable to low wages. This lowers the pre-tax reservation wage and increases search effort. A higher marginal tax rate t compresses after-tax wage gains and lowers the surplus integral, so it lowers effort. Its effect on the reservation wage can be ambiguous because it depends on how the tax schedule affects low wages relative to unemployment income.

Exercise 5.12: Search and nonhuman wealth

Each period the worker receives nonhuman income ϵ . If employed, income is

$$w + \phi\epsilon, \quad 0 < \phi < 1.$$

If unemployed, income is

$$c + \epsilon.$$

Because $\phi < 1$, nonhuman wealth is more valuable when unemployed.

If the worker accepts wage w today, the value is

$$\begin{aligned} A(w, \epsilon) &= w + \phi\epsilon + \mathbb{E} \sum_{j=1}^{\infty} \beta^j (w + \phi\epsilon_{t+j}) \\ &= w + \phi\epsilon + \frac{\beta}{1 - \beta} (w + \phi\mathbb{E}\epsilon) \\ &= \frac{w}{1 - \beta} + \phi\epsilon + \frac{\beta\phi}{1 - \beta} \mathbb{E}\epsilon. \end{aligned}$$

If she rejects,

$$R(\epsilon) = c + \epsilon + \beta Q,$$

where

$$Q = \iint V(w', \epsilon') dF(w') dG(\epsilon').$$

Thus

$$V(w, \epsilon) = \max \left\{ \frac{w}{1 - \beta} + \phi\epsilon + \frac{\beta\phi}{1 - \beta} \mathbb{E}\epsilon, c + \epsilon + \beta Q \right\}.$$

The reservation wage solves

$$\frac{\bar{w}(\epsilon)}{1 - \beta} + \phi\epsilon + \frac{\beta\phi}{1 - \beta} \mathbb{E}\epsilon = c + \epsilon + \beta Q.$$

Therefore

$$\frac{\bar{w}(\epsilon)}{1-\beta} = c + (1-\phi)\epsilon + \beta Q - \frac{\beta\phi}{1-\beta}\mathbb{E}\epsilon.$$

Since $1-\phi > 0$,

$$\bar{w}'(\epsilon) = (1-\beta)(1-\phi) > 0.$$

Higher nonhuman wealth makes unemployment relatively more attractive.

Exercise 5.13: Search and asset accumulation

The state is

$$(w, a, R, s),$$

where $s = E$ if employed and $s = U$ if unemployed. Assets cannot be negative.

For an employed worker,

$$V(w, a, R, E) = \max_{c, l, n, a'} \left\{ u(c, l) + \beta \int V(w, a', R', E) dH(R') \right\}$$

subject to

$$a' \leq R(a + wn - c), \quad l + n \leq 1, \quad a' \geq 0.$$

For an unemployed worker,

$$V(w, a, R, U) = \max \left\{ V(w, a, R, E), \max_{c, a'} \left[u(c, 1) + \beta \iint V(w', a', R', U) dF(w') dH(R') \right] \right\}$$

subject to

$$a' \leq R(a + z - c), \quad a' \geq 0.$$

The reservation wage is no longer a constant. It depends on wealth and the interest rate:

$$\bar{w} = \bar{w}(a, R).$$

Assets change the value of waiting because they help the worker self-insure during unemployment. Interest-rate risk changes the value of saving and therefore also affects the accept/reject margin.

Exercise 5.14: Temporary unemployment compensation

Let $V_1^U(w)$ be the value in the first period of an unemployment spell, when the worker receives benefit γ if she rejects. Let $V_+^U(w)$ be the value after benefits expire. Then

$$V_1^U(w) = \max \left\{ \frac{w}{1-\beta}, \gamma + \beta \int V_+^U(w') dF(w') \right\},$$

$$V_+^U(w) = \max \left\{ \frac{w}{1-\beta}, \beta \int V_+^U(w') dF(w') \right\}.$$

The reservation wages satisfy

$$\frac{\bar{w}_1}{1-\beta} = \gamma + \beta \int V_+^U(w') dF(w'),$$

$$\frac{\bar{w}_+}{1-\beta} = \beta \int V_+^U(w') dF(w').$$

Subtracting gives

$$\boxed{\bar{w}_1 - \bar{w}_+ = (1-\beta)\gamma.}$$

Thus

$$\bar{w}_1 > \bar{w}_+.$$

The hazard of leaving unemployment in state i is

$$h_i = \Pr(W \geq \bar{w}_i) = 1 - F(\bar{w}_i).$$

Since $\bar{w}_1 > \bar{w}_+$,

$$\boxed{h_1 < h_+}.$$

The worker is less likely to accept a job while benefits are available.

If employed workers can quit and requalify for benefits, define

$$Q_1 = \int V_1^U(w') dF(w'), \quad Q_+ = \int V_+^U(w') dF(w').$$

Then the employed value is

$$V^E(w) = \max \left\{ \frac{w}{1-\beta}, Q_1 \right\}.$$

The unemployment value functions are

$$V_1^U(w) = \max \{ w + \beta V^E(w), \gamma + \beta Q_+ \},$$

$$V_+^U(w) = \max \{ w + \beta V^E(w), \beta Q_+ \}.$$

The associated reservation wages satisfy

$$\bar{w}_+ = 0, \quad 0 < \bar{w}_1 \leq \bar{w}_E,$$

where

$$\bar{w}_E = (1-\beta)Q_1.$$

The key intuition is that quitting creates an option value of requalifying for benefits.

Exercise 5.15: Seasons I

In odd periods the worker receives one offer. In even periods she receives two offers. Let $V^o(w)$ and $V^e(w)$ denote values in odd and even periods. Since the best of two draws has distribution F^2 ,

$$\boxed{V^o(w) = \max \left\{ \frac{w}{1-\beta}, c + \beta \int V^e(w') dF(w')^2 \right\},}$$

$$\boxed{V^e(w) = \max \left\{ \frac{w}{1-\beta}, c + \beta \int V^o(w') dF(w') \right\}.}$$

The cutoffs satisfy

$$\frac{\bar{w}^o}{1-\beta} = c + \beta \mathbb{E}_{F^2} V^e(W'),$$

$$\frac{\bar{w}^e}{1 - \beta} = c + \beta \mathbb{E}_F V^o(W').$$

Rejecting in an odd period leads to an even period, where the worker receives two offers. Rejecting in an even period leads to an odd period, where she receives only one offer. Hence the outside option is better in odd periods, implying

$$\bar{w}^o \geq \bar{w}^e.$$

Exercise 5.16: Seasons II

Now unemployed workers always receive one offer, but employed workers face firing risk only at the start of odd periods. Let

$$V_U^e(w), \quad V_U^o(w), \quad V_E^e(w), \quad V_E^o(w)$$

be the values for unemployed/employed workers in even/odd periods.

The unemployed even-period Bellman equation is

$$V_U^e(w) = \max \left\{ w + \beta \left[\pi \int V_U^o(w') dF(w') + (1 - \pi) V_E^o(w) \right], c + \beta \int V_U^o(w') dF(w') \right\}.$$

The unemployed odd-period Bellman equation is

$$V_U^o(w) = \max \left\{ w + \beta V_E^e(w), c + \beta \int V_U^e(w') dF(w') \right\}.$$

For employed workers,

$$V_E^o(w) = w + \beta V_E^e(w),$$

$$V_E^e(w) = w + \beta \left[\pi \int V_U^o(w') dF(w') + (1 - \pi) V_E^o(w) \right].$$

Solving the two employed equations expresses V_E^e and V_E^o as affine functions of w plus constants. Substituting into the unemployed equations shows that the accept value is increasing in w while the reject value is independent of w . Thus the optimal unemployed policy is seasonal reservation wages:

$$\text{accept in even periods if } w \geq \bar{w}^e, \quad \text{accept in odd periods if } w \geq \bar{w}^o.$$

The ranking of $\bar{w}^e$ and $\bar{w}^o$ depends on the relative value of being unemployed next even versus next odd period.

Exercise 5.17: Gittins indices for beginners

There are two jobs, A and B . Job A has Markov transition matrix A_{ij} and wage $w_A(i)$. Job B has transition matrix B_{IJ} and wage $w_B(I)$.

If the worker has previously worked at both jobs, and the last observed states are i for job A and I for job B , the value is

$$V(i, I) = \max \left\{ \sum_j A_{ij} [w_A(j) + \beta V(j, I)], \sum_J B_{IJ} [w_B(J) + \beta V(i, J)] \right\}.$$

The first term is the value of choosing job A next period; the second is the value of choosing job B .

If the worker has worked only at job A , define $V_A(i)$. Then

$$V_A(i) = \max \left\{ \sum_j A_{ij} [w_A(j) + \beta V_A(j)], \sum_J \pi_B(J) [w_B(J) + \beta V(i, J)] \right\}.$$

Similarly,

$$V_B(I) = \max \left\{ \sum_j \pi_A(j) [w_A(j) + \beta V(j, I)], \sum_J B_{IJ} [w_B(J) + \beta V_B(J)] \right\}.$$

For a new worker,

$$Q = \max \left\{ \sum_i \pi_A(i) [w_A(i) + \beta V_A(i)], \sum_I \pi_B(I) [w_B(I) + \beta V_B(I)] \right\}.$$

The key idea is experimentation: trying a job gives both current wages and information about that job's future wage distribution.

Exercise 5.18: Jovanovic job matching and human capital

The worker's current wage is

$$w_t = x_t(1 - \phi_t - s_t),$$

where x_t is job-specific human capital, ϕ_t is time invested in human capital, and s_t is time spent searching. Human capital evolves as

$$x_{t+1} = G(x_t, \phi_t) = g(x_t \phi_t) - \delta x_t.$$

Search produces a new job offer μ' with probability $\pi(s_t)$.

The Bellman equation is

$$V(x) = \max_{\phi, s} \left\{ x(1 - \phi - s) + \beta \left[(1 - \pi(s))V(x') + \pi(s) \int \max\{V(\mu'), V(x')\} dF(\mu') \right] \right\},$$

where

$$x' = G(x, \phi) = g(x\phi) - \delta x.$$

Equivalently,

$$V(x) = \max_{\phi, s} \left\{ x(1 - \phi - s) + \beta V(x') + \beta \pi(s) \int [\max\{V(\mu'), V(x')\} - V(x')] dF(\mu') \right\}.$$

The worker accepts the outside offer if

$$V(\mu') \geq V(x').$$

If V is increasing, this is equivalent to

$$\mu' \geq x'.$$

For an interior optimum, the search first-order condition has the intuitive form

$$x = \beta \pi'(s) \int_{\mu': V(\mu') > V(x')} [V(\mu') - V(x')] dF(\mu').$$

The left side is the current wage cost of search time; the right side is the marginal option value of increasing the probability of a new offer.

4 Chapter 6: Recursive partial equilibrium

Exercise 6.1: A competitive firm

The firm maximizes

$$\sum_{t=0}^{\infty} \beta^t R_t,$$

where

$$R_t = p_t y_t - \frac{1}{2} d (y_{t+1} - y_t)^2.$$

Demand is

$$p_t = A_0 - A_1 Y_t,$$

and the firm believes

$$Y_{t+1} = H_0 + H_1 Y_t.$$

The Bellman equation is

$$V(y, Y) = \max_{y'} \left\{ (A_0 - A_1 Y) y - \frac{1}{2} d (y' - y)^2 + \beta V(y', Y') \right\},$$

where

$$Y' = H_0 + H_1 Y.$$

For the LQ representation, define

$$x_t = \begin{bmatrix} y_t \\ Y_t \\ 1 \end{bmatrix}, \quad u_t = y_{t+1} - y_t.$$

Then

$$x_{t+1} = A x_t + B u_t,$$

where

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & H_1 & H_0 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

The return is

$$(A_0 - A_1 Y) y - \frac{1}{2} d u^2.$$

A symmetric quadratic representation is

$$R = \begin{bmatrix} 0 & -A_1/2 & A_0/2 \\ -A_1/2 & 0 & 0 \\ A_0/2 & 0 & 0 \end{bmatrix}, \quad Q = -\frac{d}{2}.$$

Then

$$x' R x = A_0 y - A_1 y Y, \quad u' Q u = -\frac{1}{2} d u^2.$$

For the parameter values

$$A_0 = 100, \quad A_1 = .05, \quad \beta = .95, \quad d = 10, \quad H_0 = 95.5, \quad H_1 = .95,$$

the reported policy is

$$y_{t+1} = 96.9487 + y_t - 0.0463Y_t.$$

If there are n identical firms and $Y_t = ny_t$, then

$$Y_{t+1} = ny_{t+1} = n(96.9487) + (1 - n \cdot 0.0463)Y_t.$$

The Euler equation follows from the Bellman first-order condition

$$-d(y' - y) + \beta V_y(y', Y') = 0.$$

By the envelope theorem,

$$V_y(y, Y) = A_0 - A_1Y + d(y' - y).$$

Therefore

$$-d(y_{t+1} - y_t) + \beta [A_0 - A_1Y_{t+1} + d(y_{t+2} - y_{t+1})] = 0.$$

Exercise 6.2: Rational expectations equilibrium

Now the firm is representative, so in equilibrium

$$y_t = Y_t.$$

The firm takes the perceived law

$$Y_{t+1} = H_0 + H_1Y_t$$

as given when solving its problem. The solution from Exercise 6.1 has the form

$$y_{t+1} = h_0(H) + h_1(H)y_t + h_2(H)Y_t.$$

Imposing $y_t = Y_t$ gives the actual aggregate law

$$Y_{t+1} = h_0(H) + [h_1(H) + h_2(H)]Y_t.$$

A rational expectations equilibrium is a fixed point:

$$H_0 = h_0(H), \quad H_1 = h_1(H) + h_2(H).$$

Equivalently, if $M(H)$ is the aggregate law implied by optimal firm behavior under beliefs H , then

$$H = M(H).$$

Among the three candidate pairs, the reported rational expectations equilibrium is

$$(H_0, H_1) = (95.08187459215024, .95245906270392).$$

The associated feedback rule implies

$$y_{t+1} - y_t = .950819 - .0475Y_t.$$

Setting $y_t = Y_t$ gives

$$Y_{t+1} = 95.0819 + .95245Y_t.$$

A natural computational algorithm is fixed-point iteration:

$$H^{j+1} = M(H^j),$$

continued until

$$\|H^{j+1} - H^j\| < \varepsilon.$$

Exercise 6.3: Maximizing welfare

The planner maximizes discounted surplus

$$\sum_{t=0}^{\infty} \beta^t S_t,$$

where

$$S_t = \int_0^{Y_t} (A_0 - A_1 x) dx - \frac{1}{2} d(Y_{t+1} - Y_t)^2.$$

Compute the integral:

$$\int_0^Y (A_0 - A_1 x) dx = A_0 Y - \frac{A_1}{2} Y^2.$$

Thus the Bellman equation is

$$V(Y) = \max_{Y'} \left\{ A_0 Y - \frac{A_1}{2} Y^2 - \frac{1}{2} d(Y' - Y)^2 + \beta V(Y') \right\}.$$

The LQ state and control are

$$x_t = \begin{bmatrix} Y_t \\ 1 \end{bmatrix}, \quad u_t = Y_{t+1} - Y_t.$$

The law of motion is

$$x_{t+1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} x_t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u_t.$$

The payoff matrices are

$$R = \begin{bmatrix} -A_1/2 & A_0/2 \\ A_0/2 & 0 \end{bmatrix}, \quad Q = -\frac{d}{2}.$$

The Euler equation is

$$-d(Y_{t+1} - Y_t) + \beta [A_0 - A_1 Y_{t+1} + d(Y_{t+2} - Y_{t+1})] = 0.$$

For the benchmark parameters, the reported policy is

$$Y_{t+1} = 95.0819 + 0.95245 Y_t.$$

This is the same aggregate law as the rational expectations competitive equilibrium in Exercise 6.2.

Exercise 6.4: Monopoly

A monopolist chooses aggregate output Y_t and faces inverse demand

$$p_t = A_0 - A_1 Y_t.$$

Revenue is

$$p_t Y_t = (A_0 - A_1 Y_t) Y_t = A_0 Y_t - A_1 Y_t^2.$$

The Bellman equation is

$$V(Y) = \max_{Y'} \left\{ A_0 Y - A_1 Y^2 - \frac{1}{2} d(Y' - Y)^2 + \beta V(Y') \right\}.$$

Compared with the planner, the quadratic term is $-A_1 Y^2$ rather than $-(A_1/2)Y^2$. The monopolist internalizes that more output lowers the price on all units sold, so it produces less.

The LQ matrices are

$$R = \begin{bmatrix} -A_1 & A_0/2 \\ A_0/2 & 0 \end{bmatrix}, \quad Q = -\frac{d}{2}.$$

The Euler equation is

$$-d(Y_{t+1} - Y_t) + \beta [A_0 - 2A_1 Y_{t+1} + d(Y_{t+2} - Y_{t+1})] = 0.$$

For the benchmark parameters, the reported law of motion is

$$Y_{t+1} = 73.4729 + 0.9265 Y_t.$$

The steady state is

$$\bar{Y}_M = \frac{73.4729}{1 - 0.9265},$$

which is approximately half the competitive/planner steady state.

Exercise 6.5: Duopoly

There are two firms and

$$Y_t = y_{1t} + y_{2t}.$$

Firm i maximizes

$$\sum_{t=0}^{\infty} \beta^t R_{it},$$

where

$$R_{it} = p_t y_{it} - \frac{1}{2} d(y_{i,t+1} - y_{it})^2, \quad p_t = A_0 - A_1(y_{1t} + y_{2t}).$$

A Markov perfect equilibrium is a pair of value functions (V_1, V_2) and policy functions (f_1, f_2) such that, for each firm i , V_i solves the Bellman equation given the other firm's policy, and f_i attains the maximum.

Firm i 's Bellman equation is

$$V_i(y_i, y_{-i}) = \max_{y'_i} \left\{ (A_0 - A_1(y_i + y_{-i}))y_i - \frac{1}{2}d(y'_i - y_i)^2 + \beta V_i(y'_i, y'_{-i}) \right\},$$

where

$$y'_{-i} = f_{-i}(y_{-i}, y_i).$$

For the LQ game, use

$$x_t = \begin{bmatrix} y_{1t} \\ y_{2t} \\ 1 \end{bmatrix}, \quad u_{it} = y_{i,t+1} - y_{it}.$$

Firm 1's current payoff is

$$A_0 y_1 - A_1 y_1^2 - A_1 y_1 y_2 - \frac{1}{2} d u_1^2.$$

Thus one can use

$$R_1 = \begin{bmatrix} -A_1 & -A_1/2 & A_0/2 \\ -A_1/2 & 0 & 0 \\ A_0/2 & 0 & 0 \end{bmatrix}, \quad Q_1 = -\frac{d}{2}.$$

Similarly,

$$R_2 = \begin{bmatrix} 0 & -A_1/2 & 0 \\ -A_1/2 & -A_1 & A_0/2 \\ 0 & A_0/2 & 0 \end{bmatrix}, \quad Q_2 = -\frac{d}{2}.$$

For the benchmark parameters, the reported symmetric feedback rules are

$$u_{1t} = -.0878y_{1t} - .02520y_{2t} + 83.3556,$$

$$u_{2t} = -.02520y_{1t} - .0878y_{2t} + 83.3556.$$

Equivalently,

$$y_{1,t+1} = 83.3556 + .9122y_{1t} - .02520y_{2t},$$

$$y_{2,t+1} = 83.3556 - .02520y_{1t} + .9122y_{2t}.$$

The strategic effect is dynamic Cournot competition with adjustment costs: each firm's adjustment depends on both its own output and its rival's output.

Exercise 6.6: Self-control and time-inconsistent preferences

There are $T + 1$ selves, one at each date $t = 0, 1, \dots, T$. The final self has utility

$$u(c_T),$$

and self $t < T$ has preferences

$$u(c_t) + \delta \sum_{j=1}^{T-t} \beta^j u(c_{t+j}), \quad 0 < \delta \leq 1.$$

When $\delta < 1$, preferences exhibit present bias and are time inconsistent. The resource constraint is

$$k_{t+1} + c_t \leq f(k_t).$$

A Markov perfect equilibrium is a collection of value functions and policy rules such that each self optimizes given the future selves' rules.

The backward recursion uses two value functions. Let $W_j(k)$ denote the actual equilibrium continuation utility when there are j future periods left, evaluated with ordinary exponential discounting along the future equilibrium path. Start from

$$W_0(k) = u(f(k)),$$

because the last self consumes all available resources.

Given W_j , the self with $j + 1$ periods remaining solves

$$V_{j+1}(k) = \max_c \{u(c) + \beta\delta W_j(f(k) - c)\}.$$

Let the maximizer be

$$c_{j+1}(k) \in \arg \max_c \{u(c) + \beta\delta W_j(f(k) - c)\}.$$

Then update the actual continuation value as

$$W_{j+1}(k) = u(c_{j+1}(k)) + \beta W_j(f(k) - c_{j+1}(k)).$$

The difference between V and W is essential:

- V_{j+1} is the current self's objective and contains the present-bias factor δ .
- W_{j+1} records the actual equilibrium path generated by current and future selves.

Under commitment, the time-0 self controls the whole consumption plan. The sequence problem is

$$\max_{\{c_t, k_{t+1}\}_{t=0}^T} \left\{ u(c_0) + \delta \sum_{t=1}^T \beta^t u(c_t) \right\}$$

subject to

$$k_{t+1} + c_t \leq f(k_t), \quad k_0 \text{ given.}$$

A convenient dynamic representation is to define the standard exponential continuation problem

$$J_j(k) = \max_{\{c_s\}_{s=0}^j} \sum_{s=0}^j \beta^s u(c_s)$$

subject to the resource constraint. Then the time-0 commitment problem is

$$\max_{c_0} \{u(c_0) + \delta \beta J_{T-1}(f(k_0) - c_0)\}.$$

The commitment allocation generally differs from the Markov-perfect allocation because future selves cannot revise the time-0 plan.

5 Summary: the conceptual bridge

The three chapters form a sequence:

$$\boxed{\text{control} \longrightarrow \text{search and stopping} \longrightarrow \text{equilibrium consistency.}}$$

Chapter 4 teaches how to turn a dynamic problem into matrices and solve for a linear feedback rule. Chapter 5 teaches why stopping problems produce reservation rules and how outside options shift those cutoffs. Chapter 6 teaches that individual recursive decisions must aggregate into laws of motion that validate the beliefs used in optimization.